

IPv4 ADDRESSES

To uniquely identify each device on the internet, is IP Address. Ipv4 has 32-bit address (4-byte address or 4-Octect address)

Two notation are commonly used for IPv4 addresses; Binary Notation and Dotted Decimal notation.

Binary notation:

01110101 10010101 00011101 11101010

Dotted Decimal notation:

117 . 149 . 29 . 234

CLASSFUL ADDRESSING

Netid and Hostid

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0-127			
Class B	128-191			
Class C	192-223			
Class D	224-239			
Class E	240-255			

b. Dotted-decimal notation

Addresses for private network

Class	Net IDs	Blocks
A	10	1
B	172.16 ~ 172.31	16
C	192.168.0 ~ 192.168.255	256

Default masks for classful addressing

Class	Binary	Dotted-Decimal	CIDR
A	11111111 00000000 00000000 00000000	255.0.0.0	8
B	11111111 11111111 00000000 00000000	255.255.0.0	16
C	11111111 11111111 11111111 00000000	255.255.255.0	24

First Address The first address in the block can be found by setting the 32 - n rightmost bits in the binary notation of the address to 0s (using AND logic operation)

Last Address The last address in the block can be found by setting the 32 - n rightmost bits in the binary notation of the address to 1s. (using OR logic operation).

Network and Broadcast Addresses

The first address is called the network address and defines the organization itself to the rest of the world and is used by routers to direct the message sent to the organization from the outside. The last address is called Broadcast address and is used to forward a message to all nodes in the given network.

SUBNETING CLASS C ADDRESSES

In a Class C address, only 8 bits are available for defining the hosts.

Remember that subnet bits start at the left and go to the right, without skipping bits.

This means that the only Class C subnet masks can be the following:

Binary (4 th Octet)	Decimal (4 th Octet)	CIDR (for Public IP)	Number of subnets	Number of hosts per subnet	Valid subnets (4 th Octet)	Broadcast Address for each subnet
10000000	128	/25	2	128 - 2	0, 128	127, 255
11000000	192	/26	4	64 - 2	0, 64, 128, 192	63, 127, 191, 255
11100000	224	/27	8	32 - 2	Calculate yourself	
11110000	240	/28	16	16 - 2		
11111000	248	/29	32	8 - 2		
11111100	252	/30	64	4 - 2		

We can't use a /31 or /32 because we have to have at least 2 host bits for assigning IP addresses to hosts.

After choosing a possible subnet mask for the network, it is needed to determine the number of subnets, valid hosts, and broadcast addresses of a subnet that the mask provides. The questions to be answered are:
(At this point it's important that you both understand and have memorized your powers of 2.)

How many subnets? 2^x = number of subnets. x is the number of masked bits, or the 1s. For example, in 11000000, the number of ones gives us 2^2 subnets. In this example, there are 4 subnets.

How many hosts per subnet? $2^y - 2$ = number of hosts per subnet. y is the number of unmasked bits, or the 0s. For example, in 11000000, the number of zeros gives us $2^6 - 2$ hosts. In this example, there are 62 hosts per subnet. You need to subtract two for the subnet address and the broadcast address, which are not valid hosts.

What are the valid subnets? $256 - \text{subnet mask} = \text{block size}$, or increment number. An example would be $256 - 192 = 64$. The block size of a 192 mask is always 64. Start counting at zero in blocks of 64 until you reach the subnet mask value and these are your subnets.
0, 64, 128, 192.
that is, if you can count in the needed block size!

Example:

192.168.10.0 = Network address

255.255.255.128 (/25) = Subnet mask

Solution:

No. of subnets = $2^1 = 2$.

No. of hosts per subnet = $2^7 - 2 = 126$ hosts.

valid subnets address: 0 and 128, since Block Size = $256 - 128 = 128$.

Subnet Address	First Host Address	Last Host Address	Broadcast Address
192.168.0.0	192.168.0.1	192.168.0.126	192.168.0.127
192.168.0.128	192.168.0.129	192.168.0.254	192.168.0.255

SUBNETING CLASS B ADDRESSES

The Class B network address has 16 bits available for hosts addressing. This means we can use up to 14 bits for subnetting since we must leave at least two bits for host addressing. This means that the Class B subnet masks can be the following:

Binary (3 rd and 4 th Octect)	Decimal (All Octects)	CIDR (Classless Inter-Domain Routing)
10000000 00000000	255.255.128.0	/17
11000000 00000000	255.255.192.0	/18
11100000 00000000	255.255.224.0	/19
11110000 00000000	255.255.240.0	/20
11111000 00000000	255.255.248.0	/21
11111100 00000000	255.255.252.0	/22
11111110 00000000	255.255.254.0	/23
11111111 00000000	255.255.255.0	/24
11111111 10000000	255.255.255.128	/25
11111111 11000000	255.255.255.192	/26
11111111 11100000	255.255.255.224	/27
11111111 11110000	255.255.255.240	/28
11111111 11111000	255.255.255.248	/29
11111111 11111100	255.255.255.252	/30

Remember! In class B subnetting can be performed in two octets. So we have to check the block size etc. in both octets separately and then do the calculations for finding the subnet addresses and host and broadcast addresses.

Note: The Class A can be subnetted in the same way as for Class C and B. However, in that case we have subnetting possible in 3 octets as opposed to 1 or 2 subnets as in Class C or B respectively.

Practice Example #1 for Class B: 255.255.128.0 (/17)

172.16.0.0 = Network address

255.255.128.0 = Subnet mask

1_ *How many subnets?* Since 128 is 1 bit on (10000000), the answer would be 2^1 .

2_ *How many hosts per subnet?* We have 15 host bits off (10000000 00000000), so the equation would be $2^{15} - 2 = 65,534$ hosts.

3_ *What are the valid subnets?* $256 - 128 = 128$. Remember, we start at zero and count in our block size, so our subnets are 0, and 128. (For 3rd octet only. There is no subnetting in 4th octet)

Subnet Address	First Host Address	Last Host Address	Broadcast Address
172.16.0.0	172.16.0.1	172.16.127.254	172.16.127.255
172.16.128.0	172.16.128.1	172.16.255.254	172.16.255.255

Practice Example #2 for Class B: 255.255.255.128 (/25)

172.16.0.0 = Network address

255.255.255.128 = Subnet mask

Solution:

No. of *subnets* = $2^9 = 512$ subnets

No. of *hosts per subnet* = $2^7 - 2 = 126$ hosts.

valid subnets address: 0.0, 0.128, 1.0, 1.128, 2.0, ..., 255.0 and 255.128

since Block Size = $256 - 255 = 1$. (For 3rd Octet)

and Block Size = $256 - 128 = 128$. (For 4th Octet)

Subnet Address	First Host Address	Last Host Address	Broadcast Address
172.16.0.0	172.16.0.1	172.16.0.126	172.16.0.127
172.16.0.128	172.16.0.129	172.16.0.254	172.16.0.255
172.16.1.0	172.16.1.1	172.16.1.126	172.16.1.127
172.16.1.128	172.16.1.129	172.16.1.254	172.16.1.255
172.16.2.0	172.16.2.1	172.16.2.126	172.16.2.127
172.16.2.128	172.16.2.129	172.16.2.254	172.16.2.255
...	...	...	...
172.16.255.0	172.16.255.1	172.16.255.126	172.16.255.127
172.16.255.128	172.16.255.129	172.16.255.254	172.16.255.255

Implementing Variable Length Subnet Masks (VLSMs) Networks

As the name suggests, with VLSMs we can have different subnet masks for different router interfaces. For a class C network, following subnets are possible.

Prefix	Mask	Block Size	Hosts
/25	128	128	126
/26	192	64	62
/27	224	32	30
/28	240	16	14
29	248	8	6
/30	252	4	2

Keep in mind that it doesn't matter where you start your block sizes as long as you always count from zero. For example, if you had a block size of 16, you must start at 0 and count from there—0, 16, 32, 48, etc. **You can't start a block size of 16 from, say, 40 or anything other than increments of 16.**

Practice Example for Class C network using VLSM: 192.168.10.0 /24

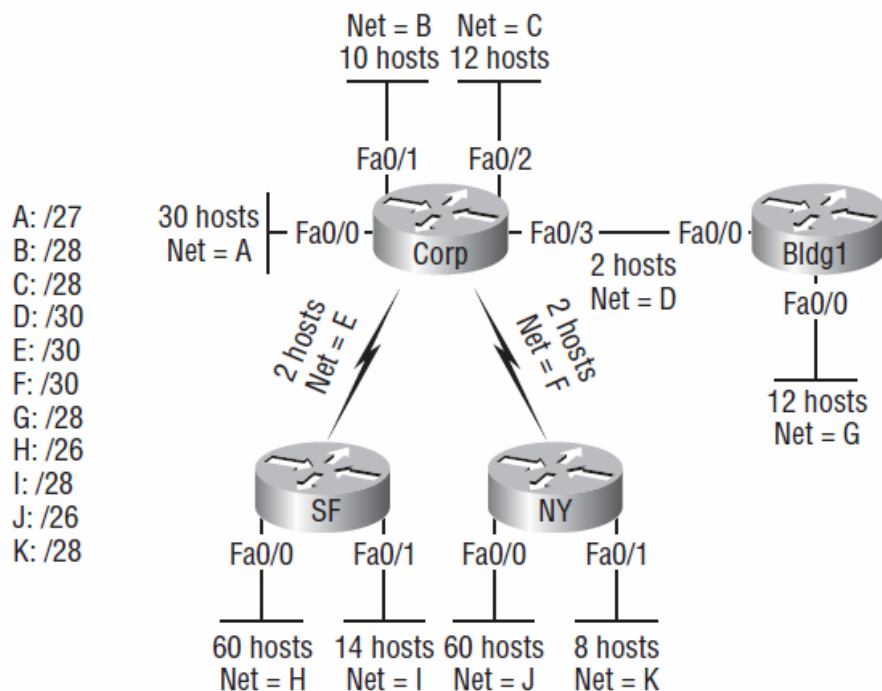


Figure shows a network with 11 networks. Create VLSM table by using block size chart to fill in the table with the subnets you need. This results in two block sizes of 64, one of 32, five of 16, and three of 4.

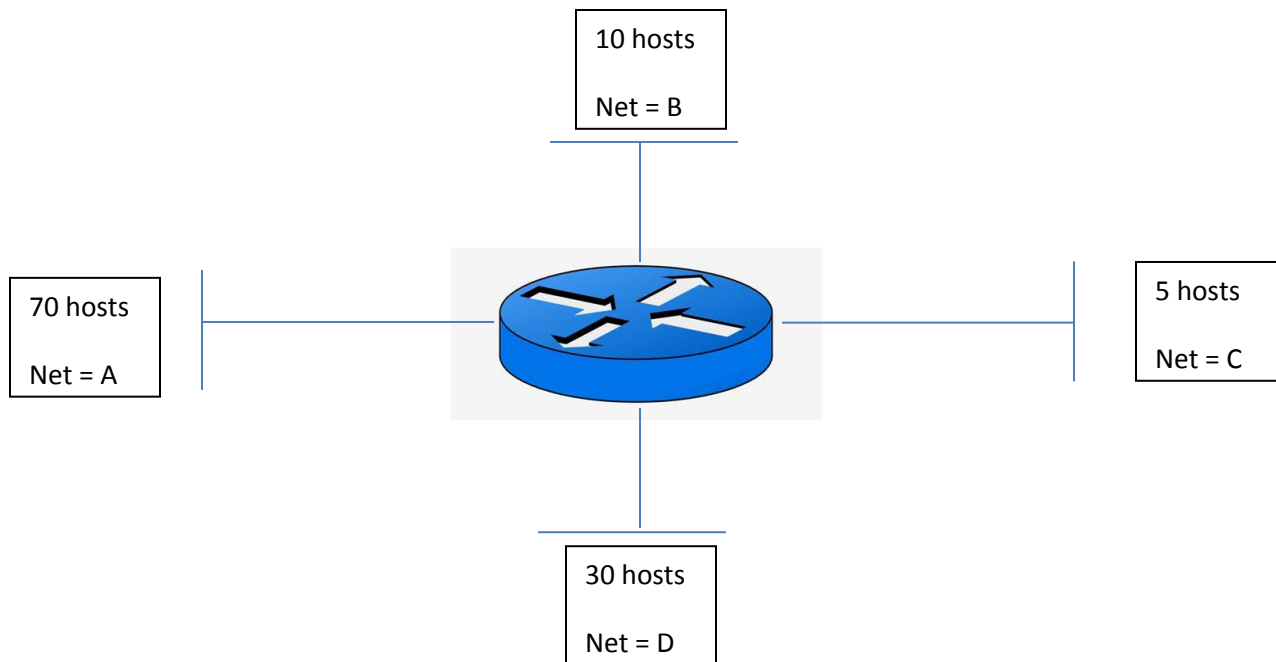
VLSM Table

Class C Network		192.168.10.0		
Network	Hosts	Block	Subnet	Mask
A	30	32	32	224
B	10	16	0	240
C	12	16	16	240
D	2	4	244	252
E	2	4	248	252
F	2	4	252	252
G	12	16	208	240
H	60	64	64	192
I	14	16	192	240
J	60	64	128	192
K	8	16	224	240

VLSM Sheet (/Chart)

0	
4	
8	
12	B - 192.16.10.0/28
16	
20	
24	C - 192.16.10.16/28
28	
32	
36	
40	
44	
48	A - 192.16.10.32/27
52	
56	
60	
64	
68	
72	
76	
80	
84	
88	
92	
96	H - 192.16.10.64/26
100	
104	
108	
112	
116	
120	
124	
128	
132	
136	
140	
144	
148	
152	
156	
160	J - 192.16.10.128/26
164	
168	
172	
176	
180	
184	
188	
192	
196	
200	I - 192.16.10.192/28
204	
208	
212	
216	G - 192.16.10.208/28
220	
224	
228	
232	K - 192.16.10.224/28
236	
240	
244	D - 192.16.10.244/30
248	E - 192.16.10.248/30
252	F - 192.16.10.252/30
256	

Practice Example #2 for Class C network using VLSM: 192.168.10.0 /24



Create VLSM table for the above four networks. You must have to assign blocks following the name sequence of subnets, that is to assign IP address block to network A first, then B, then C and finally to D.

Subnet	Hosts required	Block size assigned	Subnet Mask (in binary)	Subnet Address	Broadcast Address
A	70	128	11111111 11111111 11111111 10000000	192.168.10.0	192.168.10.127
B	10	16	11111111 11111111 11111111 11110000	192.168.10.128	192.168.10.143
C	5	8	11111111 11111111 11111111 11111000	192.168.10.144	192.168.10.141
D	30	32	11111111 11111111 11111111 11100000	192.168.10.160	192.168.10.191

The block of 32, can start only with multiple of 32 block sizes. That is, it must be either be 0, 32, 64, 96, 128, 160, 192, or 224.